

Introduction

- We solve time independent Schrodinger equation (SE),

$$H|\Phi_0\rangle = E|\Phi_0\rangle,$$

where H and E are electronic Hamiltonian and energy resp.

and $|\Phi_0\rangle$ is a one electron wave function. To account for missing instantaneous electron-electron interaction we use one of the most popular approach coupled-cluster (CC).

- CC wavefunction

$$|\Psi\rangle = e^T|\Phi_0\rangle, \quad T = T_1 + T_2 + \dots$$

with $T_n = \sum_{i_1 \dots i_n} \sum_{a_1 \dots a_n} t_{i_1 \dots i_n}^{a_1 \dots a_n} X_{i_1 \dots i_n}^{a_1 \dots a_n}$, is a cluster operator. Which

contain amplitude t and particle-hole excitation operators X , that excites electron from occupied i 's to virtual orbitals a 's.

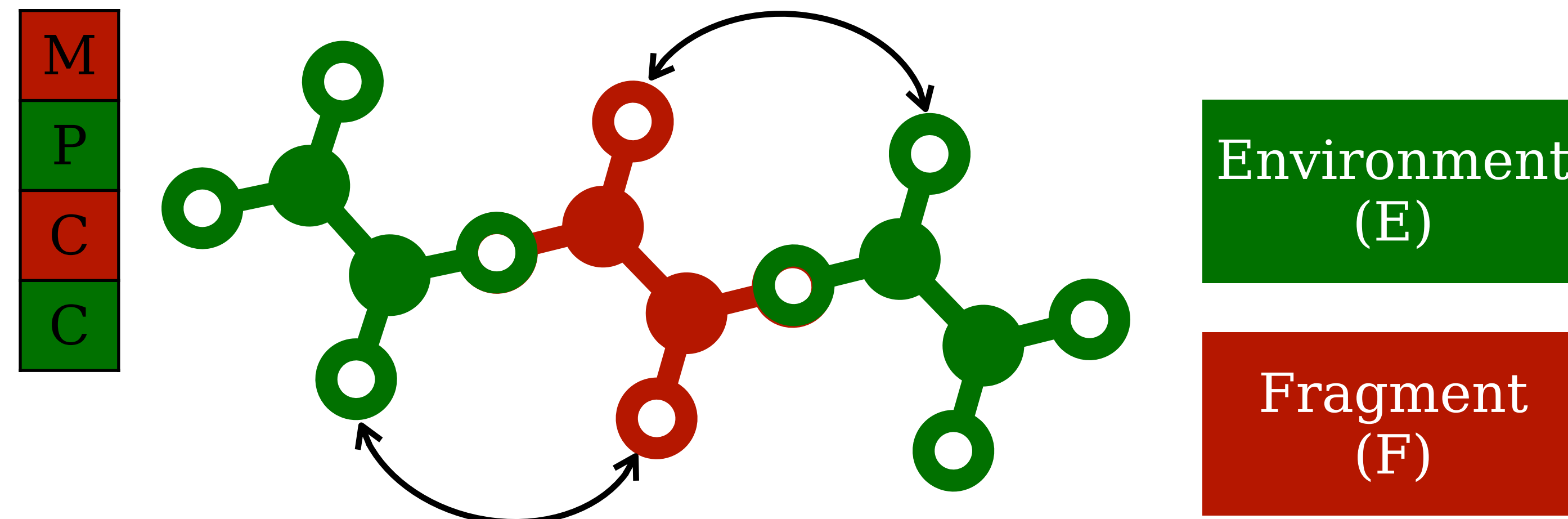
- Projecting SE to $|\Phi_0\rangle$ and excited state $|\Phi_\mu\rangle$ gives energy and amplitude respectively,

$$\langle\Phi_0|\bar{H}|\Phi_0\rangle = E, \quad \langle\Phi_\mu|\bar{H}|\Phi_0\rangle = 0, \quad \bar{H} = e^{-T}He^T$$

- Coupled-cluster theory is classified according to the highest excitation level retained in the cluster operator (singles (S), doubles (D), triples (T), ...

Methods	CC2	CCSD	CCSD(T)
Scaling	$\mathcal{O}(N^5)$	$\mathcal{O}(N^6)$	$\mathcal{O}(N^7)$

- CCSD is highly accurate but calculation for large system can become very expensive. To address this limitation, Shee et al. developed embedding approach called Moller-Plesset coupled cluster (MPCC). In MPCC, system is partitioned in to chemically important fragment treated with high-level (HL) CCSD and remaining part environment solved with low-level (LL) CC2.



- Coupled set of projective equations for LL and HL are

$$\langle\Phi_\mu^F|\bar{H}^F(\mathbf{t}^F; \mathbf{t}^E)|\Phi_0\rangle = 0, \quad \langle\Phi_\mu^E|\bar{H}^E(\mathbf{t}^E; \mathbf{t}^F)|\Phi_0\rangle = 0$$

- The energy and amplitude equations of the MPCC approach can be combined into a Lagrangian

$$\mathcal{L}(\mathbf{t}, \lambda) = \langle\Phi_0|\bar{H}|\Phi_0\rangle + \langle\Phi_0|\Lambda^F\bar{H}^F(\mathbf{t}^F; \mathbf{t}^E)|\Phi_0\rangle + \langle\Phi_0|\Lambda^E\bar{H}^E(\mathbf{t}^E; \mathbf{t}^F)|\Phi_0\rangle$$

where $\Lambda^Y = \sum_{\mu \in Y} \lambda_\mu^Y X_\mu^Y$ for $Y \in \{F, E\}$

Methodology

- As the size of the molecular system increases, the storage and manipulation of large tensors quickly become a major bottleneck in post-Hartree Fock (HF) methods. The primary computational challenge arises from the four-index two electron integral (TEI), which scale as $\mathcal{O}(N^4)$ in both storage and computational effort.

TEI with ϕ as the atomic basis function is defined as

$$G_{st}^{pq} = \iint \phi_p^*(r_1) \phi_q^*(r_2) g(r_1, r_2) \phi_s(r_1) \phi_t(r_2) dr_1 dr_2$$

with $g(r_1, r_2) \equiv \|r_1 - r_2\|_2^{-1}$.

- A standard approach to reduce the complexity of TEI tensor is Density fitting (DF), which decompose it to 2 three-index integrals.

$$G_{st}^{pq} \stackrel{\text{DF}}{\approx} \sum_{Q=1}^X J_{ps}^Q J_{qt}^Q$$

where X represents predetermined auxiliary basis set.

- DF approximation only factorize the pairs of orbital and connect them through an auxiliary basis set which imply it reduce the storage from $\mathcal{O}(N^4)$ to $\mathcal{O}(N^3)$, but overall it does not alleviate the scaling much. To address this limitation, we apply Canonical polyadic (CP) decomposition

$$J_{ps}^Q = \sum_S A_{pS} I_{sS} N_{QS}^Q$$

- Since environment constitute the largest part of the system, the LL approximation scales as $\mathcal{O}(N^5)$ and $\mathcal{O}(N^4)$ in memory, limiting the applicability of MPCC to large systems. Incorporating DF into the MPCC LL solver yields an algorithm with $\mathcal{O}(N^4)$ computational cost and $\mathcal{O}(N^3)$ storage complexity. Although DF improves scaling of LL solver, but memory requirement becomes unmanageably large for for large systems. We consider some of the terms from residual equation after applying DF to TEI.

$$\Omega_{ai} \approx - \sum_{bQj} J_{ab}^Q t_j^b J_{ji}^Q + \sum_Q J_{ai}^Q \hat{X}^Q \quad (n_o^2 n_v n_{aux} + n_o n_v n_{aux})$$

- We introduce CP approximation to each DF-TEI tensors

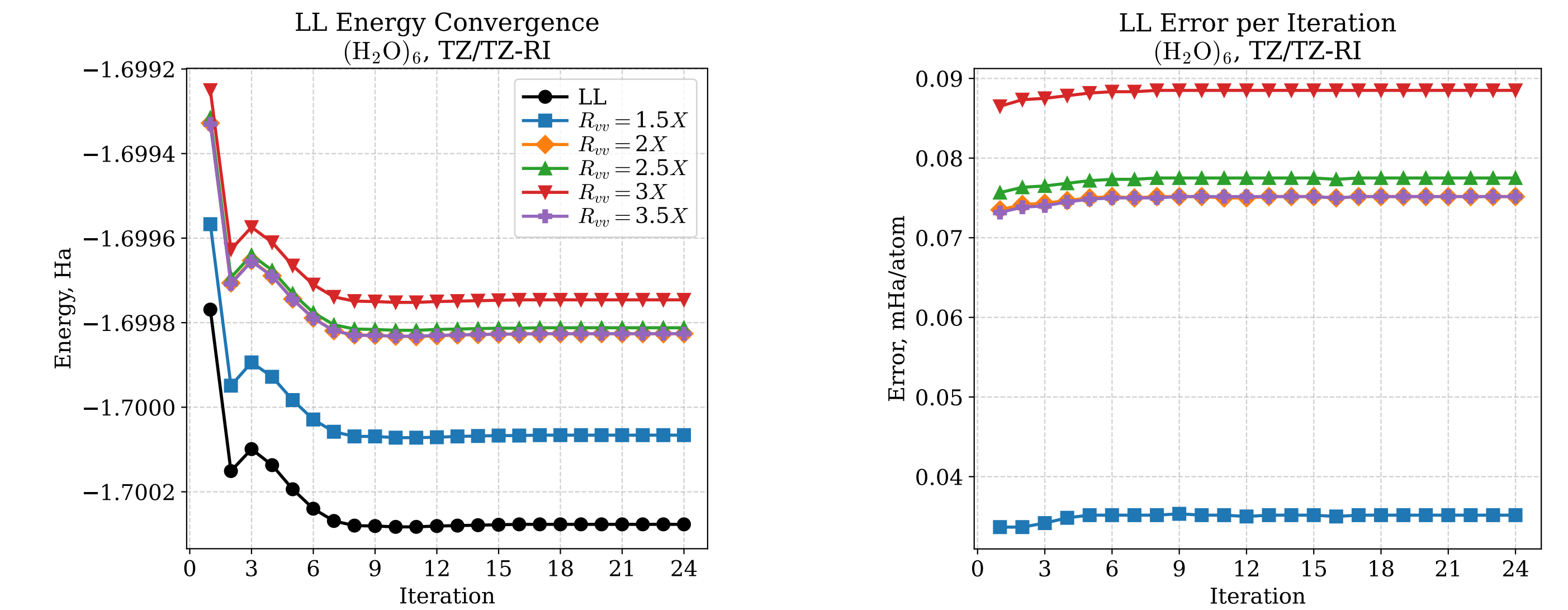
$$J_{ij}^Q \approx \sum_S I_{is} I_{js} M_{QS}, \quad J_{ab}^Q \approx \sum_S C_{as} C_{bs} V_{QS}, \quad J_{ia}^Q \approx \sum_S K_{is} A_{as} N_{QS}$$

- After CP decomposition

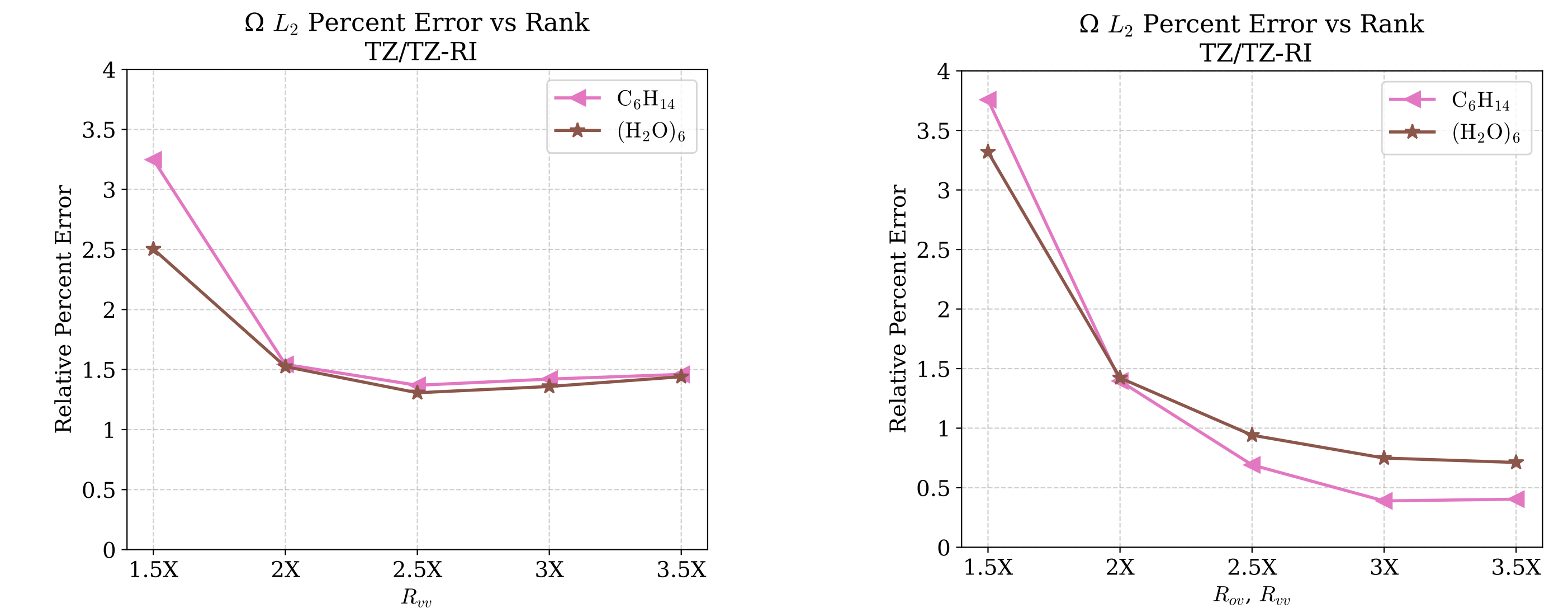
$$\Omega_{ai} \approx \sum_{ST} C_{as} [(\sum_j (\sum_b C_{bs} t_j^b) I_{jT}) (\sum_Q (R_{ab})_{QS} (P_{ij})_{QT}) I_{iT} + \sum_S C_{as} I_{is} (\sum_Q (Q_{ai})_{QS} \hat{X}^Q)] \quad \mathcal{O}(N^2 R)$$

Results

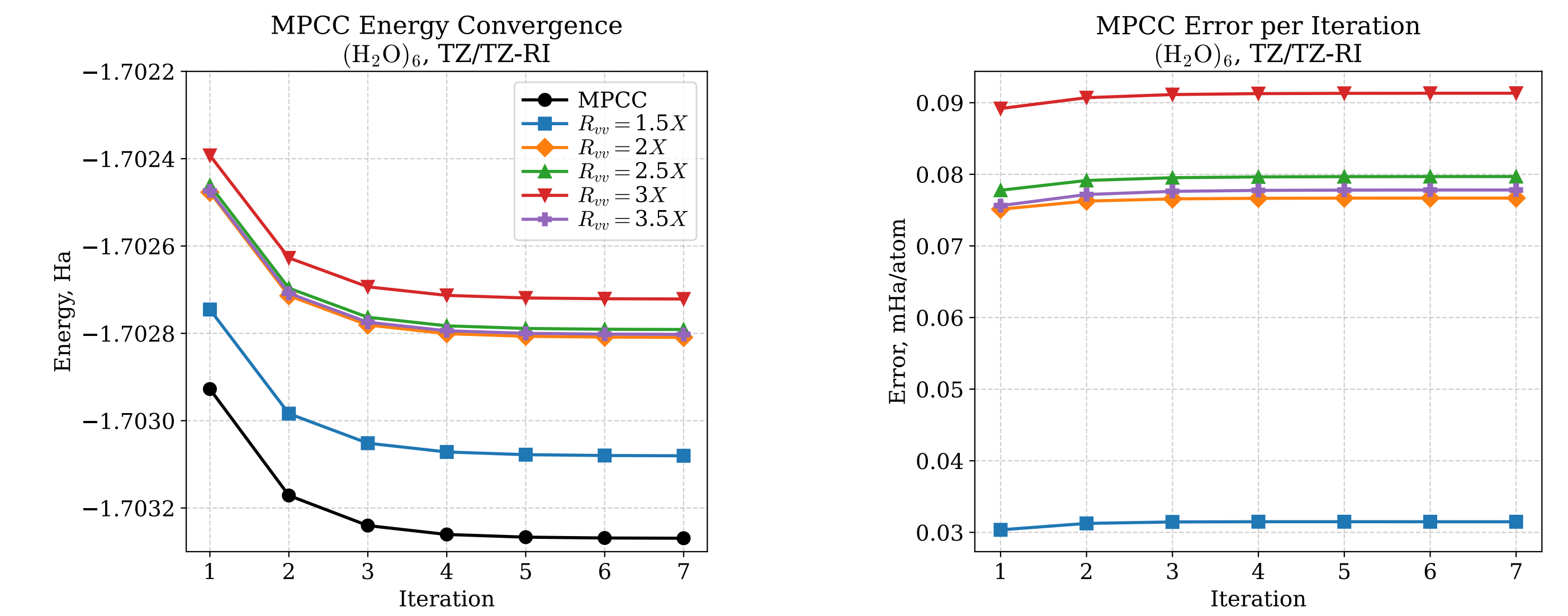
- To assess the impact of the CPD approximation within this workflow, we emulate its use by replacing the exact DF-TEI tensors with tensors reconstructed from their CPD.
- We primarily investigate two classes of molecular systems in TZ (i) Water clusters (1-6 water) (ii) Alkane chains (1-6 carbon)
- We choose ranks, $R_{oo} = X$ for J_{ij}^Q , and $R_{ov} = 2X$.
- LL convergence and error analysis



- Ω L_2 error analysis



- MPCC convergence and error analysis



- Rank scaling analysis (a), Dissociation energy (b)

